

Simulating spontaneous symmetry breaking field theory on a quantum hardware

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I. INTRODUCTION

$$H_{loc} = \sum_{j=0}^{N-1} \left(0.5\Pi_j^2 + 0.5m_I^2\Phi_j^2 + \frac{\lambda}{4!}\Phi_j^4 + f\Phi_j \right) \quad (1)$$

$$H_c = \sum_{j=0}^{N-1} \left[0.5 \sum_{e=0}^d (\Phi_{j+e} - \Phi_j)^2 + 0.5(m_0^2 - m_I^2)\Phi_j^2 \right] \quad (2)$$

$$H_s = H_{loc} + sH_c, \quad (3)$$

where $s = t/T$. Let us start with two sites so $N = 2$ and 6 qubit for each sites. The nearest-neighbor periodic condition gives

$$H_c = 0.5(\Phi_1 - \Phi_0)^2 + 0.5(m_0^2 - m_I^2)(\Phi_0^2 + \Phi_1^2) \quad (4)$$

We built single site operator using standard definitions:

$$\Phi = \frac{1}{\sqrt{2\mu}}(a + a^\dagger) \quad (5)$$

$$\Pi = -i\sqrt{\frac{\mu}{2}}(a - a^\dagger) \quad (6)$$

$$N_\Phi = 64 \quad (7)$$

Ground state of H_{loc}

$$H_{loc}^{(2)} = h_{loc}^{(0)} \otimes I + I \otimes h_{loc}^{(0)} \quad (8)$$

Diagonalization of the matrix gives

$$H_{loc}^{(2)}|\Psi_{loc}^g\rangle = E_0|\Psi_{loc}^g\rangle \quad (9)$$

II. WORKFLOW

This is what we want to do:

1. Take 2 sites, and 4 qubits for each site.
2. Find ground state of H_{loc} by classical computation and VQE: $|\psi_{loc}^g\rangle$.
3. Check the fidelity
4. Do unitary real time evolution of this ground state: $e^{-iH_s t} |\psi_{loc}^g\rangle$

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5. Find ground state of H_s for $s = 1$ with classical computation
6. Find fidelity of $e^{-iH_s t} |\psi_{loc}^g\rangle$ with ground state of H_s for $s = 1$
7. Find out those T s with the fidelity on point 6 ≥ 0.95 .
8. Vary lambda and find out the critical value of lambda for the phase change for T s from previous task.
9. Use different combination of m_0^2 and m_I^2 in the spontaneous symmetry breaking regime

III. STATUS OF WORKFLOW

A. Workflow 1

$$H_{loc} = \sum_{j=0}^1 \left(0.5\Pi_j^2 + 0.5m_I^2\Phi_j^2 + \frac{\lambda}{4!}\Phi_j^4 + f\Phi_j \right) \quad (10)$$

B. Workflow 2 and 3: Variational Ansatzs

Two different variational ansatzes are used in the VQE simulations.

C. Hardware-Efficient Ansatz (HEA)

The first ansatz consists of alternating layers of single-qubit rotations and nearest-neighbour entangling gates. The variational circuit is

$$U(\boldsymbol{\theta}) = \prod_{l=1}^L U_{CZ}^{(l)} U_{\text{rot}}^{(l)},$$

where

$$U_{\text{rot}}^{(l)} = \prod_q R_y(\theta_{q,l}) R_z(\phi_{q,l}),$$

and

$$U_{CZ}^{(l)} = \prod_{\langle i,j \rangle} CZ_{ij}.$$

This ansatz is hardware-friendly and highly expressive, but it is not directly motivated by the structure of the Hamiltonian. The entangling layer is constructed from nearest-neighbour controlled-Z gates,

$$U_{\text{ent}}^{(l)} = \prod_{\langle i,j \rangle} CZ_{ij}. \quad (11)$$

The even-odd entangling pattern mimics the connectivity of a one-dimensional lattice system. Unlike a Hamiltonian Variational Ansatz (HVA), this circuit does not directly exponentiate the Hamiltonian terms. Instead, it provides a generic parameterized quantum state with sufficient expressibility to approximate the ground state.

The VQE minimizes the expectation value

$$E(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle,$$

to approximate the ground-state energy and eigenvector of the lattice Hamiltonian.

In case of symmetry breaking regime with $m_I^2 < 0$, we have used the same HEA as described in the previous subsection. However, in the weak symmetry breaking regime,

$$m_I^2 < 0, \quad f \ll 1,$$

we observed that the VQE occasionally converged to a state with very small fidelity with the exact ground state, despite reproducing the exact ground-state energy with high accuracy. This behavior appeared only for a narrow range of coupling values λ .

The origin of this issue is the near-degeneracy of the low-energy spectrum in the weak symmetry breaking regime. For very small values of f , the scalar potential develops an approximate $\mathbb{Z}_2$ symmetry under

$$\Phi \rightarrow -\Phi,$$

leading to two nearly degenerate vacuum sectors. Consequently, the energy gap between the ground state and the first excited state becomes extremely small,

$$\Delta E = E_1 - E_0 \ll 1.$$

Since the VQE minimizes only the energy expectation value, the optimizer can converge either to the true ground state or to a nearly degenerate excited state with almost identical energy. In such cases, the energy error remains small while the fidelity with the exact ground state drops sharply.

To stabilize the optimization and consistently follow the physical vacuum branch, several improvements were implemented:

1. Warm-start continuation was introduced by using the optimized variational parameters from the previous value of λ as the initial point for the next optimization step.
2. Multiple VQE restarts with random initial conditions were performed for every value of λ .
3. An eigenstate-tracking procedure was implemented using the overlap

$$O = |\langle \psi_{\text{prev}} | \psi_{\text{trial}} \rangle|^2,$$

where ψ_{prev} is the optimized state obtained at the previous value of λ .

4. Finally, a hybrid energy-overlap selection criterion was used. Among all candidate states lying within a small energy window of the minimum-energy solution, the state with the largest overlap with the previously selected state was chosen.

This procedure successfully eliminated the spurious branch switching and allowed stable tracking of the physical vacuum state throughout the parameter scan. The observed near-degenerate structure is itself an important signature of the spontaneous symmetry breaking regime and is expected to play a significant role in the subsequent adiabatic state preparation analysis.

D. Hamiltonian Variational Ansatz (HVA)

The second ansatz is physics-inspired and constructed from generators that resemble the Pauli structure of the Hamiltonian. Each layer contains

$$e^{-i\theta_X \sum_i X_i}, \quad e^{-i\theta_Z \sum_i Z_i}, \quad e^{-i\theta_{ZZ} \sum_{\langle i,j \rangle} Z_i Z_j}.$$

The resulting variational circuit is

$$U(\boldsymbol{\theta}) = \prod_{l=1}^L e^{-i\theta_X^{(l)} H_X} e^{-i\theta_Z^{(l)} H_Z} e^{-i\theta_{ZZ}^{(l)} H_{ZZ}}.$$

Unlike the hardware-efficient ansatz, the HVA incorporates the interaction structure of the Hamiltonian directly into the circuit design. Consequently, it achieves higher fidelities with fewer layers, particularly in the symmetry-broken

phase. The results demonstrate that both hardware-efficient and Hamiltonian-inspired ansätze can reproduce the exact ground-state energy with high fidelity. However, the Hamiltonian-inspired ansatz achieves comparable accuracy with fewer layers, indicating better physics-awareness and optimization efficiency.

The symmetry-breaking term $f\phi$ modifies the vacuum structure and increases the complexity of the ground state. For small symmetry breaking ($f = 10^{-5}$), the variational states reproduce the exact ground state almost perfectly. For larger symmetry breaking ($f = 0.5$), the optimization becomes more challenging, leading to reduced fidelity, especially for the hardware-efficient ansatz.

These results illustrate the importance of ansatz design in variational quantum algorithms for lattice field theory and nonperturbative quantum many-body physics.

Feature	HEA	HVA
Physics-informed	No	Yes
Uses Hamiltonian structure	No	Yes
Generic rotations	Yes	Partly
Easier to optimize	Usually harder	Usually easier
Better for symmetry breaking	Worse	Better
Parameters needed	More	Fewer

TABLE I. Comparison between the Hardware-Efficient Ansatz (HEA) and the Hamiltonian Variational Ansatz (HVA).

In the symmetry-breaking regime,

$$m_I^2 < 0,$$

the standard Hamiltonian Variational Ansatz (HVA) initially exhibited unstable convergence and reduced fidelity with the exact ground state for certain values of the coupling λ . This issue became particularly pronounced in the weak explicit symmetry-breaking regime,

$$f \ll 1,$$

where the low-energy spectrum develops an approximate near-degeneracy associated with spontaneous symmetry breaking.

In this regime, the scalar potential possesses an approximate $\mathbb{Z}_2$ symmetry under

$$\Phi \rightarrow -\Phi,$$

leading to two nearly degenerate vacuum sectors separated by a very small energy gap,

$$\Delta E = E_1 - E_0 \ll 1.$$

As a consequence, the VQE optimization occasionally converged to different low-energy branches with nearly identical energies, resulting in small energy errors but significantly reduced fidelity with the exact ground state.

To improve the robustness of the HVA in the symmetry-breaking phase, several modifications were implemented:

1. Separate symmetry-preserving and symmetry-breaking HVA circuit constructions were introduced. In the symmetry-breaking regime, additional asymmetric variational generators were included in order to better capture the structure of the broken-symmetry vacuum sector.
2. The variational operators within each HVA layer were grouped according to the Hamiltonian decomposition, improving the representation of the competing low-energy sectors.
3. Multiple VQE restarts with different random initial conditions were performed for each value of λ in order to reduce sensitivity to local minima and near-degenerate excited states.

With these modifications, the HVA achieved stable convergence throughout the parameter scan and reproduced the exact ground-state energies with high fidelity in both the symmetry-conserving and symmetry-breaking regimes. The improved performance highlights the advantage of physics-informed variational ansätze for tracking the correct vacuum structure in systems with nearly degenerate low-energy manifolds.

VQE Ansatz	Exact ground state energy	VQE Ground state energy	Layers	Fidelity
HEA (Symmetry conserving)	1.055	1.075	8	0.9946
HVA (Symmetry conserving)	1.055	1.062	2	0.9973
HEA (Symmetry breaking, $f = 10^{-5}$)	-1.711	-1.711	8	0.99999887
HVA (Symmetry breaking, $f = 10^{-5}$)	-1.711	-1.711	6	0.9998
HEA (Symmetry breaking, $f = 0.5$)	-4.027	-4.023	8	0.9994
HVA (Symmetry breaking, $f = 0.5$)	-4.027	-4.015	6	0.9976

TABLE II. Status of ground state energy of H_{loc} and fidelity with exact ground state eigenstate for different VQEs. We have used $m_I^2 = 1$ (-1) for symmetry conserving (breaking) case. λ has been chosen to be 1 for all the cases.

1. Status of ground state

In table II, we have presented the comparison between results from classical diagonalisation and VQE for $\lambda = 1$. We have chosen $m_I^2 = 1$ (-1) for the symmetry conserving (breaking case). The fidelity is defined as

$$F_{loc} = |\langle \psi_{loc}^{VQE} | \psi_{loc}^{diag} \rangle|^2, \quad (12)$$

where $|\psi_{loc}^{diag}\rangle$ and $|\psi_{loc}^{VQE}\rangle$ are the ground state obtained by classical diagonalisation and VQE respectively. In the next step, we have considered three distinct cases:

1. Symmetric case: $m_I^2 = 1$, $f = 0$;
2. Weak Spontaneous Symmetry Breaking (Weak SSB) case: $m_I^2 = 1$, $f = 1e - 5$;
3. Strong Spontaneous Symmetry Breaking (Strong SSB) case: $m_I^2 = 1$, $f = 0.5$.

In fig. 1, we have plotted F_{loc} as a function of λ for these three cases. As we can see, for all these cases, we can achieve fidelity > 0.9965 (0.9975) with VQE when the ansatz is prepared as per HEA (HVA). In case of HVA, F_{LOC} gets a value ~ 0 for $\lambda = 0.522$ in case of weak SSB. To fix this, we adapted the same warm start, multiple starting and eigenstate-tracking system as described in section III C.

Scalar potential: The linear term tilts the potential strongly.

Physically: one vacuum becomes dominant,
the field acquires a preferred direction
the vacuum expectation value shifts

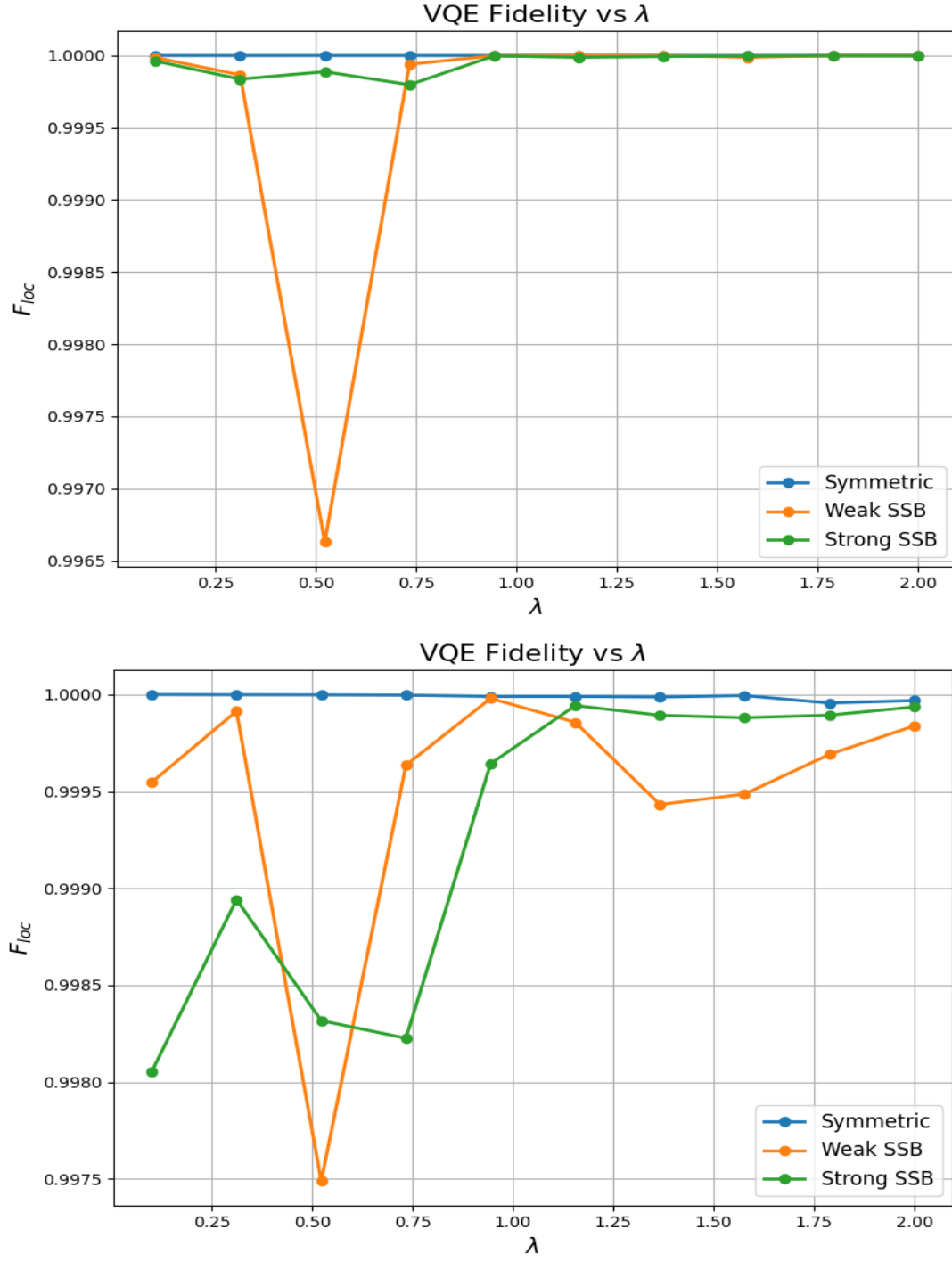


FIG. 1. F_{loc} vs λ for three distinct parameter choices of m_f^2 and f . The VQE has been performed with HEA (HVA) for the top (bottom) panel.